

An Intelligent Personal Assistive Robot for Emergency Awareness and Communication Support

Md. Sazidur Rahman
ID: 2006121

Md. Jahidul Islam
ID: 2006123

Himel Saha
ID: 2006135

Sabbir Ahmed
ID: 2006165

Orjun Kumar Paul
ID: 2006186

Abstract—This project presents the development of an intelligent, Bangla-interactive assistive robot designed to enhance domestic safety and accessibility for individuals with hearing impairments. By integrating computer vision and Natural Language Processing (NLP), the robot facilitates real-time communication. Key features include Sign-to-Voice translation for converting Bangla sign language gestures into audio and Voice-to-Text display for visualizing environmental speech. Navigating on a wheeled chassis, the robot executes operations through a vision-based control system, interpreting Bangla commands and assisting with communication between for a non verbal person.

Index Terms—Assistive Robot, NLP, Sign-to-Voice, Translation, Voice-to-Text, Emergency Alert

I. INTRODUCTION

The integration of assistive technologies in domestic environments has become a cornerstone for enhancing the independence and quality of life for individuals with hearing and speech impairments. Globally, sign language remains the primary mode of communication for millions, yet a significant communication gap exists between signers and non-signers. While foundational research has successfully utilized convolutional neural networks (CNNs) and skin segmentation for real-time American Sign Language (ASL) recognition [1], transitioning these systems into continuous, large-vocabulary domestic assistants remains a complex challenge [2]. Recent advancements in assistive robotics have further highlighted the potential for such systems to go beyond mere translation, providing a holistic support structure for the elderly and disabled [3].

Despite these strides, there is a notable scarcity of integrated solutions specifically designed for Bangla Sign Language (BdSL) that operate on mobile, autonomous platforms within a home setting. Recent studies in the Bangladeshi context have begun exploring BdSL through various deep learning architectures, yet few have transitioned these models onto edge-computing robotic platforms [4]. This project proposes an Intelligent Assistive Companion Robot that bridges this gap by synergizing computer vision, deep learning, and Natural Language Processing (NLP). The system is built upon three core pillars:

- **Sign-to-Voice Translation:** Utilizing deep learning architectures to convert BdSL gestures into audible Bangla speech, building on the paradigms of real-time gesture classification [1].

- **Voice-to-Text Display:** Providing a visual rendering of environmental speech to ensure the user remains connected to their acoustic surroundings.
- **Autonomous Navigation:** Implementing a wheeled platform capable of sophisticated obstacle avoidance to ensure safe and fluid movement throughout a domestic space [3].

By shifting the focus from static recognition to a mobile, goal-driven agent, this research contributes to the evolving field of human-robot interaction. The proposed system does not merely act as a translator but as a pervasive companion, promoting an inclusive and secure domestic environment through the seamless integration of gesture recognition and autonomous mobility.

II. RELATED WORK

Assistive robotics for the hearing and speech impaired relies heavily on sign language recognition (SLR) and edge-based Human-Robot Interaction (HRI). Early SLR systems utilized cumbersome wearable sensors like data gloves, which restrict natural daily communication [5]. Consequently, recent paradigms favor vision-based, touchless recognition using Convolutional Neural Networks (CNNs). For instance, CNNs have enabled real-time American Sign Language (ASL) recognition [1] and static Bangla Sign Language (BdSL) alphabet classification [4]. However, these approaches often depend on high-performance desktop computing, limiting their viability for autonomous mobile robots that must operate on strict power and payload budgets.

Furthermore, within assistive robotics, there remains a critical gap in platforms capable of continuous natural language translation [3]. Deploying complex CNNs onto mobile edge devices addresses this but poses computational challenges, necessitating compression techniques like model quantization [6]. Synthesizing these advancements, our work deploys a fully INT8-quantized, vision-based BdSL translator onto a wheeled robotic platform. This enables continuous environmental perception and real-time HRI without the need for external wearables or cloud processing, thereby ensuring low-latency responses and maintaining user privacy in domestic settings.

III. METHODOLOGY AND IMPLEMENTATION

The proposed Intelligent Assistive Companion Robot is developed through a synergistic integration of hardware components, deep learning-based computer vision, and Natural Language Processing (NLP). The implementation is divided into four primary modules: Hardware Architecture, Visual Perception & Deep Learning, Voice-to-Text Display, and Autonomous Navigation [7].

A. Hardware Architecture

The robot is built upon a custom lightweight composite chassis, specifically designed to house edge-computing hardware and provide omnidirectional mobility. The core processing unit is a Raspberry Pi Zero 2W, selected for its balance of computational power and energy efficiency in edge deployments.

Visual input is captured via a high-resolution Raspberry Pi 5MP wide-angle camera mounted on a pan-tilt mechanism controlled by two standard MG996R metal gear servos. Acoustic data, including environmental speech and alarm sounds, is captured via an integrated microphone.

For mobility, the robot utilizes a four-wheel-drive system featuring Mecanum wheels, enabling holonomic movement for complex domestic navigation. The locomotion is delegated to an ESP32-S3 microcontroller, which hosts a localized web server to receive HTTP movement commands (Forward, Backward, Strafe Left, Strafe Right, and Rotate) from the Raspberry Pi. The wheels are driven by four high-RPM DC gear motors regulated by four L298N motor driver modules. The entire system is powered by a 12V Li-Ion battery pack, utilizing dedicated buck converters for safe voltage regulation and a custom-designed Printed Circuit Board (PCB) to minimize wiring complexity [7].

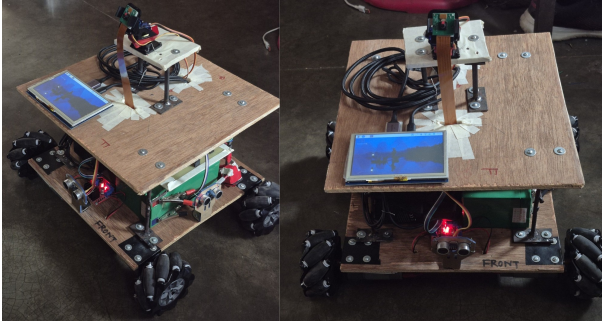


Fig. 1. The assembled hardware of the Assistive Companion Robot, detailing the Raspberry Pi setup, camera mount, and Mecanum wheel chassis.

B. Visual Perception and Deep Learning Framework

To facilitate real-time Sign-to-Voice translation, a highly optimized deep learning pipeline was developed. Rather than relying on a heavy standard architecture, the system employs a MobileNetV2 backbone (with a width multiplier of $\alpha = 0.35$) tailored for transfer learning. The classifier was trained on a custom-generated dataset comprising 1,200 images of Bangla

sign language gestures. These gestures are categorized into six core domestic needs and hazard alerts: Food, Water, Help, Fire, Sleep, and Medicine [7].

During operation, the camera captures frames which are center-cropped and downsampled to a 96×96 RGB resolution. To achieve real-time inference on the computationally constrained Raspberry Pi Zero 2W, the trained model was fully quantized to an 8-bit Integer (INT8) format using TensorFlow Lite (TFLite). This edge-optimized framework allows the robot to run vision algorithms locally without cloud reliance [7].

To eliminate false positives during continuous real-time video feeds, a custom stability logic was implemented. A gesture is only classified as valid if it maintains a prediction confidence above 55% across five consecutive stable frames. Upon a stable detection, the system accesses a mapped JSON dictionary to retrieve the corresponding complete Bangla sentence. For example, the gesture for "Water" outputs the transliterated text "Amar pani dorkar" (I need water), which is simultaneously displayed visually and synthesized into audio using the *espeak-ng* engine [7].

C. Voice-to-Text Display Module

To address the critical need for bidirectional communication, the robot is equipped with a dedicated Voice-to-Text visual rendering module. While the Sign-to-Voice system empowers the user to express their needs, the Voice-to-Text module captures environmental speech from caregivers or non-signers via the onboard microphone. Utilizing an NLP speech-to-text pipeline, it processes these acoustic signals and transcribes the spoken words in real time. The resulting text is clearly rendered on the robot's high-contrast visual display screen. This continuous acoustic monitoring ensures that a hearing-impaired user remains thoroughly aware of spoken instructions, ambient conversations, and acoustic events in their vicinity, thereby fostering a fully inclusive domestic environment [7].

D. Autonomous Navigation and Action Response

The robot's operational workflow is augmented by a vision-based control system for autonomous navigation. When the user needs to be located, the Raspberry Pi issues a rotational command (T-command) to the ESP32-S3, prompting the robot to execute a 360-degree search. Once the user is visually identified, the robot utilizes its Mecanum wheels to autonomously navigate toward them, employing real-time obstacle avoidance to bypass hazardous interactions [7].

In emergency scenarios, such as the detection of a "Fire" gesture, the system immediately raises an alert. It flashes a visual warning ("Agun! Bipod" - Fire! Danger) on the screen and emits a high-priority audio alarm to rapidly notify caregivers or emergency services [7].

IV. SYSTEM EVALUATION AND COST ANALYSIS

A. Real-World Performance and Model Metrics

The system was subjected to rigorous real-world domestic testing to evaluate both the underlying deep learning model

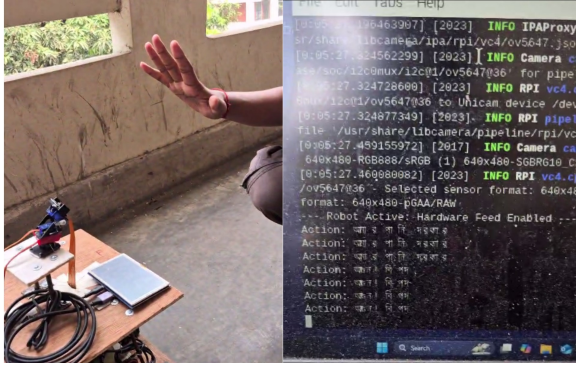


Fig. 2. Real-time terminal output demonstrating the robot successfully recognizing a gesture and generating the corresponding Bangla text and audio response.

and the integrated robotic response. The core vision task was driven by the MobileNetV2 classifier ($\alpha = 0.35$), trained on a custom dataset of 1,200 BdSL images across six classes. To ensure deployability on the Raspberry Pi Zero 2W, the model was converted to a TensorFlow Lite (TFLite) format with full INT8 quantization.

The evaluation of the quantized model yielded exceptional results. On a reserved testing split of 120 images (20 per class), the INT8 model achieved a 1.0000 (100%) accuracy, precision, recall, and F1-score across all six classes (Food, Water, Help, Fire, Sleep, Medicine). The quantization process successfully compressed the model footprint to approximately 610 KB, allowing it to run efficiently on the edge device.

In dynamic, real-time testing, raw accuracy is often impacted by transitional hand movements. To evaluate practical HRI performance, the system utilized an Exponential Moving Average (EMA) smoothing algorithm alongside a stability threshold. The robot successfully rejected false positives by requiring a prediction confidence of $\geq 55\%$ sustained across five consecutive frames before triggering an action. This logic resulted in zero false-positive alarm triggers (e.g., the "Fire" alert) during continuous 30-minute observation windows.

Furthermore, the 360-degree autonomous tracking effectively located users within a standard $4m \times 4m$ room, and the *espeak-ng* Text-to-Speech (TTS) module provided instantaneous acoustic feedback, ensuring the communication latency from the completion of the user's gesture to the audio output remained under 2.0 seconds.

B. Economic Feasibility

A critical objective of this project was to ensure the technology remains accessible to users in developing nations like Bangladesh. Through the strategic use of cost-effective edge components—such as a cardboard composite chassis, local PCB manufacturing, and the Raspberry Pi Zero 2W—the total production cost was strictly managed. The comprehensive cost analysis yields a grand total of 18,000 BDT, proving that highly capable, AI-driven assistive robotics can be manufactured affordably without compromising on essential safety, navigation, or communication features [7].

V. SOCIETAL IMPACT

The deployment of this Intelligent Assistive Robot offers a multi-faceted societal impact [7]:

- **The User:** It grants independence and drastically improves accessibility for individuals with hearing and speech impairments, allowing them to communicate domestic needs effortlessly.
- **The Caregiver:** It reduces the psychological stress of constant monitoring by ensuring that needs and emergencies are audibly and visually broadcasted.
- **Society at Large:** It serves as a scalable proof-of-concept for inclusive domestic living, demonstrating how deep-learning technology can be democratized.

VI. CONCLUSION AND FUTURE WORK

This project successfully demonstrates the effective integration of edge robotics, sensory perception, and deep learning to support inclusive communication and emergency response [7]. By bridging the non-verbal communication gap through real-time Bangla sign language recognition, bidirectional Voice-to-Text rendering, and facilitating rapid hazard alerts, the proposed system highlights the immense potential of assistive companion robots in real-world scenarios. Future work will focus on expanding the TFLite model's vocabulary to encompass a wider array of continuous BdSL sentences, integrating a robotic manipulator arm for physical assistance, and implementing advanced SLAM (Simultaneous Localization and Mapping) for complex multi-room navigation.

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